



INAI

Intelligent Network for Accessible & Inclusive Maternal Healthcare

PRESENTED BY:

Pretty Pixels



https://github.com/DIYA210408/INAI_by_pretty_pixels

PROBLEM STATEMENT AND OBJECTIVE

Pregnant women in remote and underserved communities often face limited access to timely prenatal care and emergency support due to long distances, poor connectivity, and limited digital literacy.

Existing healthcare solutions often depend on smartphones, internet access, text messages, or one-way communication, making them difficult to use for women with basic phones or limited literacy.

As a result, pregnancy danger signs may go unrecognized until the condition becomes critical, leading to delays in seeking appropriate medical care.





Our Solution

- Voice-First Healthcare: Enables mothers to communicate their symptoms naturally through a two-way voice conversation.
- No Smartphone or Internet Required: Accessible through a regular phone call, making it suitable for women using basic mobile phones.
- AI-Powered Triage: Listens to symptoms, asks relevant follow-up questions, and categorizes cases based on risk level.
- Localized Communication: Designed to understand regional languages, dialects, informal speech, and unstructured responses.



TECHONOLGIES USED

Backend Implementation

- FastAPI – Core backend & APIs
- Twilio Voice – Handles incoming calls
- Speech Processing – Captures & converts voice to text
- AI Endpoint – Symptom extraction & follow-up questions
- Voice Guidance – Sends responses via Twilio
- Testing – Uvicorn + ngrok

Flow

Phone Call → Twilio → FastAPI → Speech → Risk Analysis → Voice Response

AI and SPEECH

- STT: Voice → Text
- API Keys : Language Correction
- Symptom Extraction: Identifies symptoms
- Conversation Logic: Generates follow-up questions
- TTS: AI response → Voice

Flow

Patient Voice → STT → AI → Symptoms → Follow-up → TTS → Patient



TECHONOLGIES USED

HEALTHCARE WORKERS DASHBOARD

- React – Dashboard interface
- Backend APIs – Patient & alert data
- Risk Filters – High / Moderate / Low
- Case Management – Review & update cases

Flow

Database → API → Dashboard → Risk Filter → Review → Update

ALERTS

- Risk Engine: Clinical rule-based assessment
- Database: Stores patient cases & assessments
- Alert System: Flags high-risk cases
- Worker Mapping: Assigns cases to healthcare workers

Flow

Symptoms → Clinical Rules → Risk Level → Database → Alert

WORKFLOW

1. CALL INITIATION

Pregnant mother calls the helpline using any basic mobile phone.



2. VOICE INPUT

Twilio receives the call and captures the mother's spoken response.



3. SPEECH-TO-TEXT

Speech recognition converts the mother's voice into text for AI processing.



4. AI SYMPTOM ANALYSIS

GPT analyzes the reported symptoms and identifies potential pregnancy danger signs.



5. CONVERSATIONAL FOLLOW-UP

The AI asks relevant follow-up questions based on the mother's responses.



6. RISK CLASSIFICATION

The system categorizes the case as:
HIGH RISK | MODERATE RISK | LOW RISK



7. VOICE GUIDANCE

The AI generates a simple response and converts it into speech for the mother.



8. ALERT & RECORD

High-risk cases are stored in the database and trigger an alert for the designated healthcare worker.

